

BME 140 W24

Lab #3 Report

Lab: B1-2

Operational Amplifiers

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Due: 2/15/24

Introduction

The laboratory experiment aims to explore the fundamental characteristics and applications of operational amplifiers (op-amps). Through the design and construction of four distinct circuits, namely the voltage follower, inverting amplifier, non-inverting amplifier, and summing circuit, students will gain insights into the behavior and functionality of op-amp-based systems. Understanding op-amps is crucial in various fields including biomedical electronics, where precise signal processing and amplification are essential for applications such as medical instrumentation and signal conditioning. By investigating concepts like input/output impedance, gain, linearity, and saturation characteristics, students will develop proficiency in circuit analysis and design, paving the way for advanced electronic applications in biomedical engineering and beyond.

Theoretical Background and Design

A) Voltage Follower

Gain	R_1	R_2
$\frac{V_{out}}{V_{in}} = 1$	10 k Ω	10 k Ω

The voltage divider comprises two 10k Ω resistors situated between V_{in} and ground. With a gain of 1, the output voltage aligns with the input voltage due to their series connection, resulting in equivalence between the output and input voltages in an open circuit. Since the circuit lacks a load, it doesn't draw any current from the op-amp input. V_{in} is applied at the positive input terminal and is mirrored at the negative input. The input voltage V_{in} is 5V.

B) Inverting Amplifier (Conditions: $G = -10$, $R_1 = 10$ k Ω)

Gain	R_1	R_2
$\frac{V_{out}}{V_{in}} = \frac{-R_2}{R_1} = -10$	10 k Ω	100 k Ω

The inverting amplifier, with a gain of -10, comprises resistors $R_1 = 10$ k Ω and $R_2 = 100$ k Ω arranged in series. V_{out} is derived from the negative input of the op-amp, resulting in a reduced output impedance. The positive input terminal of the op-amp is linked to the power supply.

C) Non-Inverting Amplifier (Conditions: $G = +10$)

Gain	R_1	R_2
$\frac{V_{out}}{V_{in}} = 1 + \frac{R_2}{R_1} = +10$	1 k Ω	9 k Ω

The non-inverting amplifier, designed with a gain of 10, utilized resistors with the values $R_1 = 1$ k Ω and $R_2 = 9$ k Ω . Due to the unavailability of a 9 k Ω resistor, a 2 k Ω and an 18 k Ω resistor were substituted, maintaining the equivalent ratio. One end of the resistor was grounded while the other end was linked to the negative input terminal of the op-amp, with the resistors connected to the output terminal.

D) Summing Circuit (Conditions: $V_o = -10(V_1 + 2V_2)$)

Gain	R_1	R_2	R_f
$V_o = -(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2)$	10 k Ω	5 k Ω	100 k Ω

The summing circuit utilized resistors with the following specifications: $R_1 = 10 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$, and $R_f = 100 \text{ k}\Omega$. R_1 and R_f were positioned in parallel relative to each other and linked to the negative input terminal of the op-amp. The negative terminal of the op-amp hosted one extremity of the R_f resistor, while the other extremity was connected to the op-amp's output terminal.

Results and Discussion

A) Voltage Follower

Load Resistor R_L	V_{out1} (Open Circuit)	$V_{out1} (R_L)$	V_{out2} (Follower) (Open Circuit)	$V_{inB} (R_L)$	$V_{out2} (R_L)$
1 k Ω	$2.46 \pm 0.01 \text{ V}$	$410.4 \pm 0.1 \text{ mV}$	$2.472 \pm 0.001 \text{ V}$	$2.459 \pm 0.001 \text{ V}$	$2.472 \pm 0.001 \text{ V}$
10 k Ω	<i>Same voltage</i>	$1.638 \pm 0.001 \text{ V}$	<i>Same voltage</i>	$2.458 \pm 0.001 \text{ V}$	$2.472 \pm 0.001 \text{ V}$
100 k Ω	<i>Same voltage</i>	$2.431 \pm 0.001 \text{ V}$	<i>Same voltage</i>	$2.458 \pm 0.001 \text{ V}$	$2.472 \pm 0.001 \text{ V}$
1 M Ω	<i>Same voltage</i>	$2.446 \pm 0.001 \text{ V}$	<i>Same voltage</i>	$2.458 \pm 0.001 \text{ V}$	$2.472 \pm 0.001 \text{ V}$

a. *Does the buffer circuit help?*

The buffer does help as we can see from the table above, in the case where there isn't a voltage follower, the output voltage depends on the load resistor. Therefore, if the load resistance increases the voltage drops. In the case with a voltage follower is equivalent to the input voltage, and therefore the voltage doesn't drop as the load resistor increases.

b. *What is the gain of the voltage follower?*

The voltage gain is 1 because the output voltage follows the input voltage.

c. *What is the function of the voltage follower?*

The function of the voltage follower is to provide approximately the same voltage to a load, acting as a buffer, by not producing amplification or attenuation to the signal.

d. *What is the input impedance and output impedance of the voltage follower?
Assume an ideal Op-Amp.*

The ideal Op-Amp is assumed thus the input impedance of the voltage follower will be much larger, with extraordinarily little current drawn from the circuit. The output impedance is exceptionally low.

B) Inverting Amplifier

a) Plot V_{out} vs V_{in}

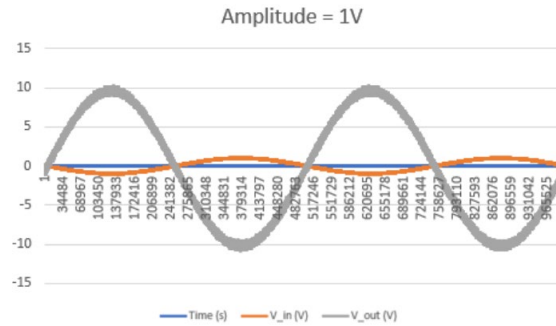


Figure 1: Inverting Amplifier Sinusoidal Wave with Amplitude of 1V

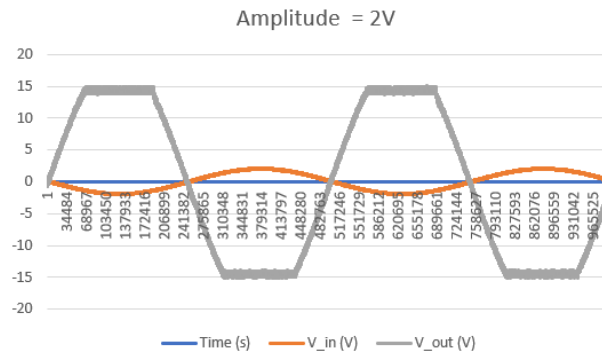


Figure 2: Inverting Amplifier Sinusoidal Wave with Amplitude of 2V

b) What is the input voltage range over which the circuit behaves linearly?

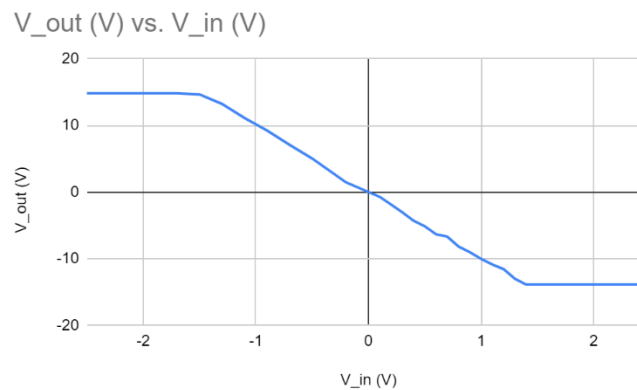


Figure 3: Variable Power Supply (V_{in} vs V_{out})

The input voltage range is between $V_{in} = -1.7$ and $V_{in} = 1.4$

c) What is the maximum and minimum output voltage possible?

The Maximum Output = 14.82V and the minimum output = -13.86V.

- d) *What is the gain of the circuit from the V_{out} vs V_{in} measurement in the linear region (use the slope of the curve to determine the gain)? Does it agree with your prediction? Is the error between the predicted value and measured value within the accuracy of the resistor used?*

$$\text{Gain} = \frac{(-13.86 \pm 0.01) - (14.82 \pm 0.01) \text{ V}}{1.5 - (-1.5) \text{ V}} = -9.56 \pm 0.2\%$$

$$\% \text{ error} = \left| \frac{-10 - (-9.56)}{-10} \right| * 100 = 4.4\%$$

Note: This is within the $\pm 5\%$ tolerance of our resistors

From the measurement in procedure 3, plot the data recorded. Did you see an opposite phase between input and output? Why is the waveform for one of the input sinusoidal signals not sinusoidal anymore?

An opposite phase between the input and output is seen the output signal is negative, inverting amplifier flipped the sign of the input signal. The data obtained for the amplitude of 2V for the sinusoidal signal is not correct due to inputting the wrong voltage into the functional generator. If done correctly, the second sinusoidal figure should show a saturated input waveform, since the input is peak to peak. The input voltage at 2V would be 20V, where the theoretical maximum voltage is 15, thus making it saturated.

- a. *What change(s) you will make in the circuit if you want a gain of 100 instead of 10?*

If the gain was to change to 100, then we need to increase the resistance of R_2 to equal $1\text{M}\Omega$.

C) Noninverting Amplifier

- a. Plot V_{in} vs V_{out}

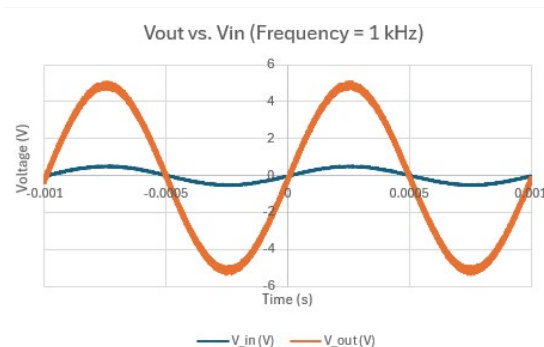


Figure 4: Noninverting Amplifier Sinusoidal Wave with Frequency of 1kHz (Non-Saturated)

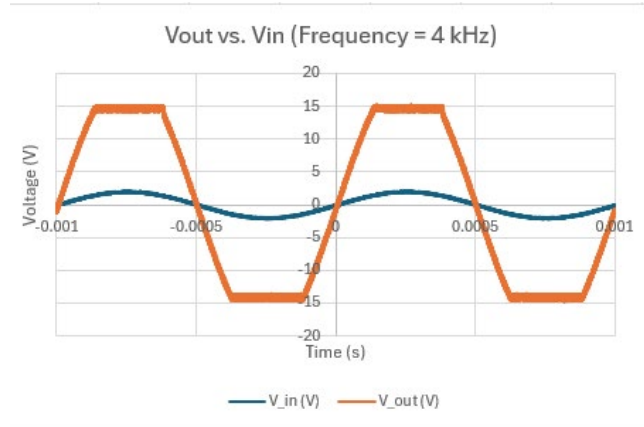


Figure 5: Noninverting Amplifier Sinusoidal Wave with Frequency of 4kHz (Saturated)

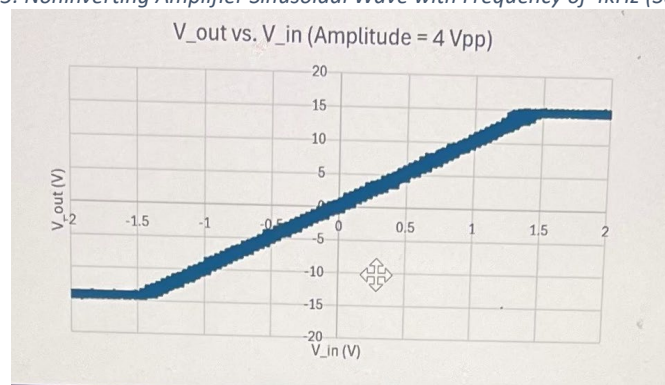


Figure 6: Vin vs Vout (borrowed from lab partner as Excel kept crashing to generate this)

b. What is the input voltage range over which the circuit behaves linearly?

Between -1.4V and 1.4V.

c. What is the maximum and minimum output voltage possible?

The Maximum Output = 14.5V and the minimum output = -14.4V.

d. What is the gain of the circuit from the Vout vs Vin measurement in the linear region (use the slope of the curve to determine the gain)?

$$\text{Gain} = \frac{(-14.4 \pm 0.01) - (14.5 \pm 0.01) \text{ V}}{1.4 - (-1.4) \text{ V}} = -10.32 \pm 0.2\%$$

$$\% \text{ error} = \left| \frac{-10 - (-10.32)}{-10} \right| * 100 = 3.2\%$$

e. From the measurement in procedure 3, plot the data recorded. Did you see an opposite phase between input and output? Why?

An opposite phase between input and output is not seen, because the output gain becomes positive since the input voltage signal is directly connected to the non-inverting terminal.

f. What is the input resistance? Assume an ideal op-amp for this noninverting amplifier circuit.

The input resistance is $10\text{k}\Omega$.

D) Summing Circuit

a. Plot data acquired in procedures 3 and 4.

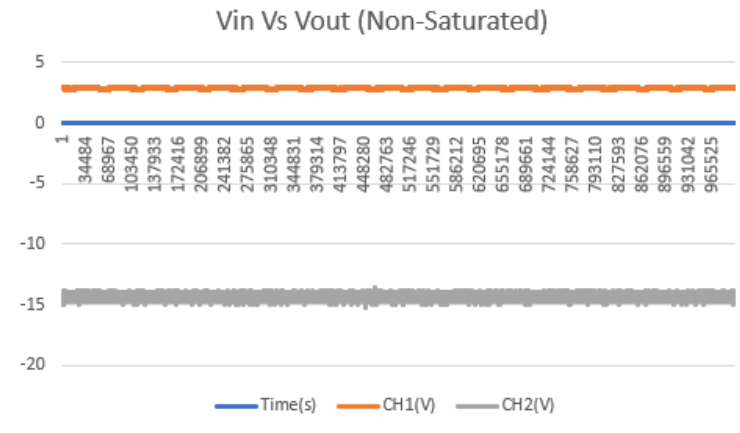


Figure 7: Summing Amplifier Sinusoidal Wave (Non-Saturated)

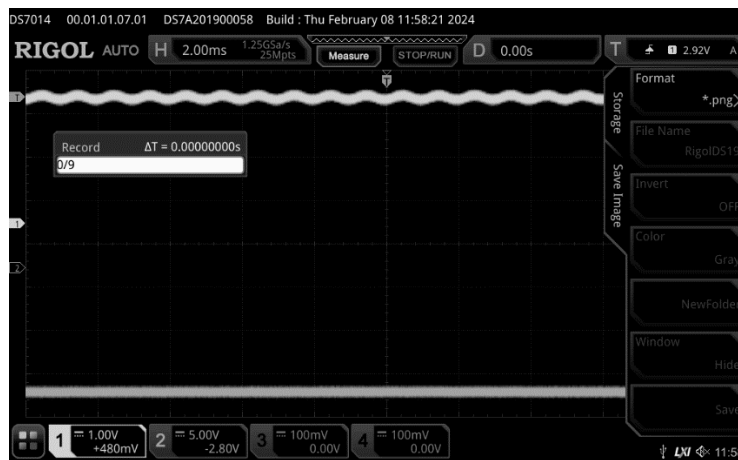


Figure 8: Summing Amplifier from Rigol (Correctly Oscillating at $y=0$)

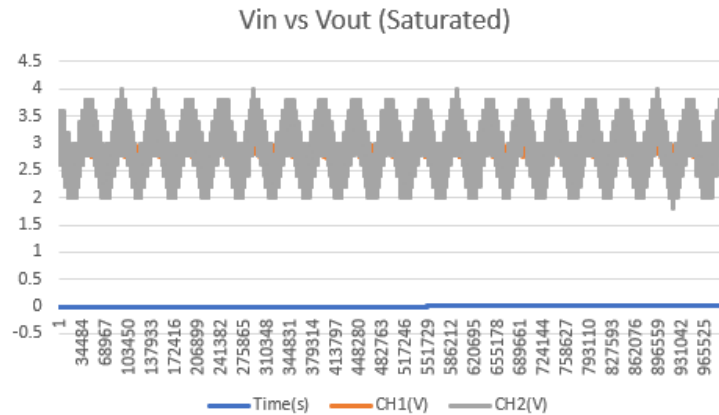


Figure 9: Summing Amplifier (Saturated)

b. Why V_{out} measured in procedures 3 and 4 are different?

The voltage out is measured differently because there is an offset of 3V in procedure 3 and V_2 was grounded (0V).

c. What is the relation between the offset voltage of V_1 and the DC voltage of V_2 ? Does the measured V_2 agree with the predicted value?

$V_2 = -0.5 V_1$, therefore, to get the zero-offset bias V_2 should be around -1.5V. The measured value of V_2 , which was recorded through the digital multimeter, was found to be -1.621 ± 0.001 V. That's relatively close with a percent error of -8.067%.

d. What is the function of V_2 ? Describe the case where this circuit is useful.

The function of V_2 is to cancel out the offset from the input. This circuit is useful in the case of multiple signals passing through the circuit.

Conclusion

The main concepts in this lab are how the op-amp operates via designing and building the voltage follower, inverting amplifier, non-inverting amplifier, and summing circuit and how implementing the ideal op-amp law can help simplify circuits. For instance, an op-amp can be used to create a positive feedback loop that feeds a voltage that will be a corresponding phase with the input voltage, as well as facilitate a negative feedback loop that feeds a voltage that can be out of phase with that of the input voltage. A conceptual aspect that I have learned from this lab is the Golden Rules of the ideal op-amp, which can be summarized into two rules. The first is that the potential voltage difference between the negative and positive terminal of the op-amp is so small that it can be assumed to be zero. The second rule states that no current flows into the input terminals. Finally, when using the golden rules, it's important to consider the gain of the circuit to obtain the correct resistors needed to build the circuit.